

Introduction to Relation

Lecture-8

Lattice



Lattice

Introduction

A partially ordered set in which every pair of elements has both a least upper bound and a greatest lower bound is called lattice.

- The least upper bound is known as **join** ($\vee$).
- The greatest lower bound is known as **meet** ($\wedge$).

Lattice

Count.....

Ex. Is the poset $(\mathbb{Z}^+, |)$ a lattice?

Sol. Let a and b be two positive integers.

The least upper bound of two integers are the L.C.M. and the greatest lower bound of two integers are the greatest common divisor.

$\Rightarrow$ this poset is a lattice.

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Determine whether the poset $(\{1, 2, 3, 4, 5\}, |)$ and $(\{1, 2, 4, 8, 16\}, |)$ are lattices.

Determine whether $(P(S), \subseteq)$ is a lattice where S is a set.

Lattice

Count.....

Theorem : For any $a, b \in L$ where L is a lattice, $a \leq a \vee b$
And $a \wedge b \leq a$

Let L be a nonempty set closed under two binary operations called meet($\wedge$) and join($\vee$).

Then L is called a lattice if it satisfies:

L1: Commutative Law

$$a \wedge b = b \wedge a \quad \text{and} \quad a \vee b = b \vee a$$

L2: Associative Law

$$(a \wedge b) \wedge c = a \wedge (b \wedge c) \quad \text{and} \\ (a \vee b) \vee c = a \vee (b \vee c)$$

Let L be a nonempty set closed under two binary operations called meet($\wedge$) and join($\vee$).

Then L is called a lattice if it satisfies:

L3: Absorption Law

$$a \wedge (a \vee b) = a \quad \text{and} \quad a \vee (a \wedge b) = a$$

where a, b, c are elements in L

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Duality

The dual of any statement in a lattice is the statement obtained by interchanging meet and join.

Ex. The dual of $a \wedge (b \vee a) = a \vee a$
is $a \vee (b \wedge a) = a \wedge a$

Lattice

Count.....

Theorem : For any Lattice $a, b \in L, a \leq a \vee b$ and $a \wedge b \leq a$

Proof: Since $a \vee b$ is the Upper bound of a, b
 $\therefore a \leq a \vee b$

Similarly $a \wedge b$ is the Lower bound of a, b
 $\therefore a \wedge b \leq a$

Theorem : For any Lattice $a, b, c, d \in L$, if $a \leq b$ and $c \leq d$ then $a \vee c \leq b \vee d$ and $a \wedge c \leq b \wedge d$.

Proof: Given that $a \leq b$ and $c \leq d$, we know that $b \vee d$ is the upper bound of $b, d \therefore b \leq b \vee d$ and $d \leq b \vee d$

$$\Rightarrow a \leq b \vee d \text{ and } c \leq b \vee d$$

$\Rightarrow b \vee d$ is the upper bound of a, c

But $a \vee c$ is the least upper bound of a, c

$$\therefore a \vee c \leq b \vee d$$

Lattice

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Theorem : For any Lattice $a, b, c, d \in L$, if $a \leq b$ and $c \leq d$ then $a \vee c \leq b \vee d$ and $a \wedge c \leq b \wedge d$.

Proof: Given that $a \leq b$ and $c \leq d$, we know that $a \wedge c$ is the lower bound of a, c $\therefore a \wedge c \leq a$ and $a \wedge c \leq c$

$\Rightarrow a \wedge c \leq b$ and $a \wedge c \leq d$

$\Rightarrow a \wedge c$ is the lower bound of b and d

$\Rightarrow \therefore a \wedge c \leq b \wedge d$

Show that the operations of meet and join on a lattice is commutative

$$a \vee b = b \vee a \quad \text{and} \quad a \wedge b = b \wedge a$$

Proof:

$$\begin{aligned} a \wedge b \\ &= \text{glb } \{a, b\} = \text{glb } \{b, a\} \\ &= b \wedge a \end{aligned}$$

$$\begin{aligned} a \vee b \\ &= \text{lub } \{a, b\} = \text{lub } \{b, a\} \\ &= b \vee a \end{aligned}$$

Show that the operations of meet and join on a lattice is idempotent $a \wedge a = a$ & $a \vee a = a$

Proof : $a \wedge a$
= glb $\{a, a\}$ = glb $\{a\}$ = a
 $a \vee a$
= lub $\{a, a\}$ = lub $\{a\}$ = a

Ex. Show that the operations of meet and join on a lattice is associative

$$(a \wedge b) \wedge c = a \wedge (b \wedge c)$$

and $(a \vee b) \vee c = a \vee (b \vee c)$

Proof:

Since $(a \vee b) \vee c = \text{lub } \{a \vee b, c\}$

$$\Rightarrow a \vee b \preceq (a \vee b) \vee c \quad \text{.....(1)}$$

$$\text{and} \quad c \preceq (a \vee b) \vee c \quad \text{.....(2)}$$

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Since $a \vee b = \text{lub } \{a, b\}$

$\therefore a \preceq a \vee b$ (3)

and $b \preceq a \vee b$ (4)

From (1) & (3),

$a \preceq (a \vee b) \vee c$ [by transitivity](5)

From (1) & (4),

$b \preceq (a \vee b) \vee c$ [by transitivity](6)

From (2) & (6),

$$b \vee c \leq (a \vee b) \vee c$$

[by definition of join](7)

From (5) & (7),

$$a \vee (b \vee c) \leq (a \vee b) \vee c$$

[by definition of join](8)

Also, $a \leq a \vee (b \vee c)$ (9)

$$b \leq b \vee c \leq a \vee (b \vee c) \quad \text{.....(10)}$$

$$c \leq b \vee c \leq a \vee (b \vee c) \quad \text{.....(11)}$$

From (9) & (10),

$$a \vee b \leq a \vee (b \vee c) \quad \text{.....(12)}$$

From (11) & (12),

$$(a \vee b) \vee c \leq a \vee (b \vee c) \quad \text{.....(13)}$$

From (8) & (13),

$$(a \vee b) \vee c = a \vee (b \vee c) \quad [\text{by anti-symmetry}]$$

Lattice

Count.....

Show that the operations of meet and join on a lattice holds
Absorption law

$$a \wedge (a \vee b) = a \quad \text{and} \quad a \vee (a \wedge b) = a$$

Proof :

Since $a \wedge b = \text{glb } \{a, b\}$

$$a \wedge b \leq a \quad \text{.....(1)}$$

$$\text{Also,} \quad a \leq a \quad \text{.....(2)}$$

From (1) & (2),

$$a \vee (a \wedge b) \leq a \quad \text{.....(3)}$$

Also $a \leq a \vee (a \wedge b) \quad \text{.....(4)}$

[by definition of lub]

From (3) and (4),

$$a \vee (a \wedge b) = a$$

Lattice

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Sub lattice

Let S be nonempty subset of a lattice L . Then S is called a sub lattice of L if S itself is a lattice.

Example 1: If S be the set and $P(S), \subseteq$ then only those collections $P(S)$ are sub lattices of L which are closed under both operation $\cup, \cap$. If $S = \{a, b, c, d\}$ then $\{\{\emptyset\}, \{a\}, \{a, c\}, \{c\}, \{a, b, c\}\}$ would be sub lattice but $\{\{\emptyset\}, \{a\}, \{b\}, \{c\}, \{b, c\}\}$ would not form a sub lattices as $\{a\} \cup \{c\} = \{a, c\}$ is not belong to set.

Lattice

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Sub lattice

Let S be nonempty subset of a lattice L . Then S is called a sub lattice of L if S itself is a lattice.

Example 2: If $L = D(30)$ having the hasse diagram then $D(6)$, $D(10)$, $D(15)$ are sub lattices.

In general if m/n then $D(m)$ is sub lattice of $D(n)$.

Lattice

Count.....

Distributive Lattice:

A lattice L is said to be **distributive** if it satisfies the distributive laws i.e. if $a, b, c \in L$ then

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

and

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

Otherwise, L is said to be non-distributive.

Theorem : In any distributive lattice,
 $a \wedge x = a \wedge y$ and $a \vee x = a \vee y$
together imply $x = y$.

Proof: $x = x \vee (x \wedge a) = x \vee (y \wedge a)$
 $= (x \vee y) \wedge (x \vee a)$
 $= (y \vee x) \wedge (y \vee a)$
 $= y \vee (x \wedge a) = y \vee (y \wedge a) = y$

Lattice

Count.....

Bounded Lattices: A Lattice L is called bounded if L has both a lower bound, denoted by 0 such that $0 \leq x, \forall x \in L$ and an upper bound, denoted by 1 such that $x \leq 1, \forall x \in L$

The bounds 0 and 1 are called universal bounds.

In bounded lattice,

$$a \vee 1 = 1 \quad a \wedge 1 = a$$

$$a \vee 0 = a \quad a \wedge 0 = 0$$

for any element a in L .

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Theorem : Every finite lattice L is bounded.

Proof: Let $L = \{a_1, a_2, \dots, a_n\}$ is a finite lattice then
 $a_1 \wedge a_2, \dots, \wedge a_n$ and $a_1 \vee a_2, \dots, \vee a_n$ are lower and upper
bound of L .

Thus, the lower and power bounds of L exists.
 $\therefore$ every finite lattice is bounded.

Lattice

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Ex. Let $L = \{1, 2, 3, 6\}$. Then $(L, |)$ is a bounded lattice. In this lattice, lower bound is 1 and upper bound is 6.

EX. The lattice $(\mathbb{N}, \leq)$, where $\mathbb{N}$ is the set of natural numbers and $\leq$ is the 'less than or equal to' relation on $\mathbb{N}$, is not bounded because this lattice has no upper bound. Although $(\mathbb{N}, \leq)$ has lower bound namely 1.

Complement

Let L be a bounded lattice with lower bound 0 and upper bound 1 . Let a be any element of L .

An element x in L is called a complement of a if

$$a \vee x = 1 \quad \text{and} \quad a \wedge x = 0$$

Complement of $0 = 1$

Complement of $1 = 0$

Complements need not exist and need not be unique.

Complemented Lattice

A lattice L is said to be complemented if L is bounded and every element in L has a complement.

Lattice

Count.....

Ex. The lattice $(P(S), \subseteq)$ of the power set of any set S is complemented. In this lattice each element has unique complement.

Lattice

Count.....

Theorem : Prove that, in a distributive lattice, if an element has a complement then this complement is unique.

Let $a \in L$ be any element and let us assume that a_1, a_2 be two distinct complements of a .

We know, by the definition of complement

$$a \vee a_1 = 1 \quad ; \quad a \wedge a_1 = 0$$

$$a \vee a_2 = 1 \quad ; \quad a \wedge a_2 = 0$$

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$$\text{Now } a_1 = a_1 \vee 0 = a_1 \vee (a \wedge a_2) = (a_1 \vee a) \wedge (a_1 \vee a_2) = 1 \wedge (a_1 \vee a_2) = a_1 \vee a_2$$

$$a_1 = a_1 \vee a_2$$

$$\text{Now } a_2 = a_2 \vee 0 = a_2 \vee (a \wedge a_1) = (a_2 \vee a) \wedge (a_2 \vee a_1) = 1 \wedge (a_1 \vee a_2) = a_1 \vee a_2$$

$$a_1 = a_2$$

Lattice

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Complete Lattice

A lattice L is complete if every subset of L has a glb and lub

- A complete lattice is always bounded but a bounded lattice need not be a complete lattice
- Every finite lattice is complete.

Lattice

Count.....

Ex. Consider the lattice $L = \{0, 1, 2, 3, 6\}$ under divisibility relation. Then $(L, |)$ is complete lattice.

Modular Lattice

A lattice L is modular if

$$a \leq c \Rightarrow a \vee (b \wedge c) = (a \vee b) \wedge c$$

Ex. Every distributive lattice is modular.

Proof: let $(L, \leq)$ be a distributive lattice and $a, b, c \in L$ be such that $a \leq c$. Thus if $a \leq c$ then $a \vee c = c$.

$$\begin{aligned} \text{Now } a \vee (b \wedge c) &= (a \vee b) \wedge (a \vee c) \\ &= (a \vee b) \wedge c \end{aligned}$$

Theorem: if $(L, \vee, \wedge)$ is distributive complemented lattice, then Demorgan,s laws.

$$(a \vee b)' = a' \wedge b'$$

and $(a \wedge b)' = a' \vee b'$

Proof: Let L is bounded by definition of a complement lattice. To proof $(a \vee b)' = a' \wedge b'$

We simply prove that $(a' \wedge b')$ is complement of $(a \vee b)$ that is

$$(a \vee b) \vee (a' \wedge b') = 1$$

$$(a \vee b) \wedge (a' \wedge b') = 0$$

$$\begin{aligned}\text{We have } (a \vee b) \vee (a' \wedge b') &= [(a \vee b) \vee a'] \wedge [(a \vee b) \vee b'] \\ &= [(a \vee a') \vee b] \wedge [a \vee (b \vee b')] \\ &= (1 \vee b) \wedge (a \vee 1) \\ &= 1 \wedge 1 \\ \therefore (a \vee b) \vee (a' \wedge b') &= 1\end{aligned}$$

$$\begin{aligned}\text{We have } (a \vee b) \wedge (a' \wedge b') &= [(a \wedge (a' \wedge b'))] \vee [b \wedge (a' \wedge b')] \\ &= [(a \wedge a') \wedge b'] \vee [a' \wedge (b \wedge b')] \\ &= (0 \wedge b') \vee (a' \vee 0) \\ &= 0 \vee 0 \\ \therefore (a \vee b) \wedge (a' \wedge b') &= 0\end{aligned}$$

Homomorphism

Let L and M be lattices. A mapping $f : L \rightarrow M$ is called homomorphism if it satisfies

$$f(x \vee y) = f(x) \vee f(y)$$

$$f(x \wedge y) = f(x) \wedge f(y)$$

Isomorphism

Let L and M be lattices. A mapping $f : L \rightarrow M$ is called isomorphism if it satisfies –

- i) f is one-to-one
- ii) f is onto
- iii) f is homomorphism